Brian K. Fung

PharmD, MPH, CNA

Clinician-Engineer · Founder · Clinical AI, Informatics, Interoperability & Public Policy

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Profile

Entrepreneurial clinical AI product builder and technical product leader with 13+ years taking complex products, platforms, and workflows from ambiguous problems to launch. Residency-trained informatics pharmacist who contributed to the electronic prescribing sections of the ONC's 21st Century Cures Act Final Rule, modernized enterprise clinical systems across 80 hospitals and clinics at Mayo Clinic, and served as Staff Clinical Informatics Data Architect at Verily; now a founder, full-stack engineer, and certified nursing assistant who shipped multiple 0-to-1 AI products in the past year, including haau3, a CMS-recognized consumer caregiving platform among the first seven apps in the CMS Medicare App Library. Four years applying AI to product development and evaluation across multilingual conversational AI, retrieval-augmented generation, workflow automation, clinical decision support, and frontier-model benchmarking, with deep roots in health-data standards and interoperability. Equally fluent in clinical and technical domains: translates ambiguous clinical needs into precise specifications, data models, and shipped software, and technical tradeoffs back into terms clinical teams can act on.

Education

Stanford Healthcare AI Leadership & Strategy (HAILS) May 2027 – Jun 2027
Stanford University School of Medicine *Stanford, CA*

A hybrid Stanford executive program for healthcare leaders focused on AI innovation, implementation, and strategy.

Graduate Certificate, Biomedical Data Science 2022 – 2023
Stanford University School of Medicine *Stanford, CA*

Hands-on model development across machine learning, deep learning, and large language models, including training, fine-tuning, and evaluation.

Master of Public Health 2018 – 2020
Johns Hopkins Bloomberg School of Public Health *Baltimore, MD*

Doctor of Pharmacy 2009 – 2013
University of Florida *Gainesville, FL*

Bachelor of Science, Human Nutrition 2005 – 2009
University of Florida *Gainesville, FL*

Postgraduate Training

PGY-2 Pharmacy Informatics Resident Jul 2014 – Jun 2015
University of Utah Health *Salt Lake City, UT*

- Led cross-hospital medication workflow and security initiatives across seven hospitals, including an enterprise authentication transition affecting 3,500+ nurses.

PGY-1 Pharmacy Practice Resident Jun 2013 – Jun 2014
Sarasota Memorial Hospital *Sarasota, FL*

- Led EHR training for a 40+ FTE pharmacy department during conversion from McKesson Horizon Meds Manager to Allscripts while maintaining clinical pharmacist responsibilities.

Licensure & Certifications

Certified Nursing Assistant (CNA) — Jan 2026 – 2028
Florida Department of Health

FHIR Foundation Implementer — Jul 2024
FHIR Foundation

HL7 FHIR R4 Proficiency Certificate — Apr 2024
HL7 International

Mayo Quality Fellow, Silver — 2018
Mayo Clinic

Mayo Quality Fellow, Bronze — 2018
Mayo Clinic

Board Certified Pharmacotherapy Specialist (BCPS) — 2016 – 2023
Board of Pharmacy Specialties

Epic Certifications — 2014 – 2025
Epic Systems

Willow Inpatient · Willow Inpatient Graduate · Inpatient Clinical Decision Support · Bugsy Infection Control · Reporting Workbench · BestPractice Advisories, Basic Badge · Alternatives Badge · Usability Badge

Making a Difference in Infectious Disease Pharmacotherapy (MAD-ID) — 2014

Pharmacist Licensure — 2013 – Present

Minnesota · Utah · Florida

Academic Appointments

Clinical Assistant Professor — Sep 2016 – Present
University of Florida College of Pharmacy — *Gainesville, FL*

Instructor of Pharmacy — 2019 – 2022
Mayo Clinic College of Medicine and Science — *Rochester, MN*

Professional Experience

Founder & AI Product Lead — Jun 2025 – Present
Brian K Fung Media, LLC — *Brandon, FL*

Independent product work, started during a period of full-time family caregiving, which is where haau3 originated.

- Conceived and built haau3, a consumer caregiving platform that unifies longitudinal health records and helps patients and caregivers securely access, understand, and share health information; sole builder from concept through general availability, setting the roadmap across identity, interoperability, conversational AI, and caregiver workflows.
- Took haau3 to general availability in the CMS Health Tech Ecosystem for both the Conversational AI and Kill the Clipboard use cases, recognized by CMS at the program's first-anniversary event in July 2026; among the first seven apps listed in the CMS Medicare App Library, sourcing records through a TEFCA QHIN and CMS Aligned Network.
- Built and iterated Cora, a multilingual text- and voice-based conversational AI assistant grounded in patient-authorized records; combined Gemini Live, RAG, deterministic knowledge retrieval, and source-linked explanations, evolving the UX from text to speech-to-text to speech-to-speech based on caregiver feedback.

- Launched Unscripted Rx from 0-to-1 as a pharmacy workforce and recruiting discovery product with 30,000+ connected entities; grew it organically to 10,000+ monthly views with top-10 Google search visibility and built AI-assisted job-posting ingestion and content normalization to reduce manual population and improve coverage.
- Built medlistIQ, an auditable medication-reconciliation API that normalizes and deduplicates fragmented medication data; owned product concept, UX, data model, architecture, and implementation end to end.
- Advise CEOs, CTOs, CPOs, and product and engineering leaders on AI-enabled product strategy, platform architecture, and EHR-integrated workflows.

Staff Clinical Informatics Data Architect Sep 2024 – Jun 2025

Verily *Fairfax, VA*

Promoted to Staff after earning the top company performance rating (5 of 5) in the preceding review cycle; rated significantly above expectations in three of six cycles overall and recognized with a 2024 Verily Values Award.

- Directed enterprise FHIR governance as an appointed member of Verily's FHIR Council, defining 200+ implementation guidance artifacts, semantic models, review patterns, and developer tooling, and leading a company-wide proposal for FHIR proficiency assessments.
- Designed evaluation frameworks comparing deterministic clinical reasoning with frontier LLM approaches for AI-assisted type 2 diabetes decision support, clarifying accuracy, explainability, safety, and implementation tradeoffs to guide Verily's clinical AI strategy.
- Shaped strategic direction for EHR integration through SMART on FHIR, pharmacy management, and externalization of Verily's care platforms, translating digitized care pathways into requirements for productized platform APIs.
- Launched Verily's first medical-coding enrichment modules, standardizing procedure (HCPCS) and immunization coding across products, by building a FHIR terminology service that combined BigQuery with a FHIR proxy to an enterprise terminology-management platform, alongside FHIR terminology data-quality measures.
- Led harmonization and enrichment of third-party health information exchange data, turning an ambiguous company initiative into a scoped delivery roadmap by mapping C-CDA-to-FHIR transformation complexity and educating cross-functional teams on the tradeoffs.
- Advanced a new market opportunity with a value-based care provider by digitizing its clinical care pathways, mapping concepts to FHIR value sets and configuring executable BPM+ models, starting with heart failure and expanding to additional pathways.
- Delivered the medical-device shipment backend within Verily's device-integration initiative, extending a unified device data model across first- and third-party devices in support of connected-care programs.
- Influenced engineering and product practice through internal publications on EHR integration, Epic data models, device data models, FHIRPath and CQL, and applications of LLMs, establishing shared collaboration channels between informatics and partner teams.

Senior Clinical Informatics Data Architect Apr 2022 – Sep 2024

Verily *Fairfax, VA*

Partnered across product, engineering, UX, research, clinical, legal, and privacy teams to translate ambiguous user and clinical needs into implementation-ready architecture, earning 10+ cross-team peer bonuses and mentoring pharmacy informatics interns and early-career engineers.

- Drove the Q1 2024 launch of Lightpath for type 1 diabetes with Highmark, earning a director spot bonus and an engineering peer bonus, by resolving launch-critical issues and automating device fulfillment from CRM through FHIR to BigQuery.
- Led semantic and workflow architecture using BPM+ and FHIR, producing 100+ CMMN, BPMN, DMN, and data artifacts that connected user journeys and clinical logic to APIs, decisions, and data contracts across all care teams.

- Shipped the clinical content behind the Verily Value Suite MVP, a flagship 2022 launch, by modeling 140+ Mayo Clinic heart-failure (HFrEF) pathway concepts into executable decision models and authoring the company's BPM+ modeling style guide.
- Established Verily's enterprise terminology-management capability by leading vendor analysis and selection of a clinical terminology and content-management platform, co-leading its implementation, and launching terminology extracts into BigQuery.
- Scaled care-pathway authoring beyond the informatics team by enlisting 20+ clinical data managers as authors, fully automating QA/QC of new models with a CI/CD pipeline, and creating FHIR and BPM+ curricula, including 14 tutorial videos among the company's most-viewed internal documentation.
- Standardized race and ethnicity data capture across Verily against federal standards and contributed key designs for laboratory results and error handling, care surveys, and professional task workflows adopted by all care teams.
- Partnered with data science teams throughout as the clinical and informatics subject-matter expert to build, train, and evaluate machine-learning and deep-learning applications across care programs, shaping training data, clinical validity, and FHIR-aligned model outputs.
- Designed Onduo's medication-review pipeline with data science and engineering, authoring the specification mapping model outputs to FHIR and proposing therapeutic-class-to-NDC eligibility mappings using a commercial drug compendium, aligned to ADA 2023 guidelines.
- Prototyped executable DMN and BPMN clinical decision models, spanning cardiac assessment, KCCQ-12 scoring, and an end-to-end CMS Enhanced Oncology Model flow, that guided Verily's strategic selection of care-pathway tooling.
- Organized and mapped 145 clinical concepts from the Mayo care-pathway program to UMLS, controlled vocabularies, and HL7 FHIR within the first months of joining, reconciling Mayo's existing decision models to eliminate duplicated mapping work.
- Served as Verily's pharmacist subject-matter expert for medications, pharmacy, and clinical workflows, consulting across product areas and delivering pharmacy cost-savings analyses, including IV-to-PO conversion opportunities for cardiac surgery populations.

Clinical Informatics Pharmacist Jun 2015 – Apr 2022

Mayo Clinic *Rochester, MN*

- Led end-to-end product and workflow transformation across four Epic module implementations, covering antimicrobial stewardship, infection control, clinical decision support, and Willow pharmacy, spanning 80 hospitals and clinics within a $1.5B enterprise program, from discovery and requirements through validation, training, rollout, and iteration.
- Modernized a 25+ year rules-based antimicrobial monitoring platform, converting 75+ custom legacy rules from CBAM and TheraDoc into Epic through 2.5 years of system discovery, workflow redesign, decision-support configuration, testing, and production deployment.
- Built NHSN regulatory reporting for CAUTI, CLABSI, SSI, VAE, and LabID events, plus internal surveillance reporting for bloodstream infections, carbapenem-resistant organisms, and multidrug-resistant organisms, with a distribution strategy designed for low ongoing maintenance.
- Designed physician-facing reminder alerts for catheter-associated urinary tract infection, central line-associated bloodstream infection, and umbilical catheters, and coordinated the enterprise prevention workflows behind them.
- Enabled bidirectional laboratory interoperability between Mayo's SOFT laboratory system and two of the largest external reference labs, Quest Diagnostics and Labcorp, integrating 10,000+ external microbiology records by mapping laboratory tests, organisms, and susceptibilities with LOINC and SNOMED CT.
- Converged an enterprise list of indications for all systemic antimicrobials and built them into Epic, alongside restricted medication workflows using alternatives, order questions, best practice advisories, and scoring.

Graduate Student Intern Jun 2019 – May 2020

Office of the National Coordinator for Health IT *Washington, DC*

MPH practicum and capstone placement, concurrent with Mayo Clinic. Practicum supervisor: Debbie Bucci. Capstone advisor: Thomas Mason, MD.

- Assisted in drafting the electronic prescribing sections for the ONC's Cures Act Final Rule, fulfilling provisions of the 21st Century Cures Act.
- Assisted in summarizing public comments on the ONC's Cures Act proposed rule during the notice of proposed rulemaking for electronic prescribing.
- Analyzed the implications of the 21st Century Cures Act on clinician burden attributable to health information technology and electronic health records, work that became the MPH capstone.
- Coordinated a one-day national stakeholder symposium on patient matching for prescription drug monitoring programs.

Pharmacy Intern Jun 2011 – Aug 2011

Malcom Randall Veterans Affairs Medical Center *Gainesville, FL*

Pharmacy Intern Jun 2010 – Aug 2010

Walgreens *Boynton Beach, FL*

Publications

1. Lessard S, Zimmermann E, Patel L, **Fung B**, Rivera CG. Putting words into action: adopting vancomycin infusion reaction terminology. *American Journal of Health-System Pharmacy*. 2023;80(4):175-176. doi:10.1093/ajhp/zxac345
2. Franky, **Fung BK**. Patient centered view through the lens of a pharmacist. In: Ball MJ, ed. *Nursing Informatics: A Health Informatics, Interprofessional and Global Perspective*. 5th ed. New York: Springer; 2022.

Research Projects

Phantom Codes: benchmarking hallucination, hierarchical accuracy, abstention, robustness, and cost across 24 LLM and prompting configurations for medical concept normalization May 2026

Independent research; open evaluation harness released publicly, with retrieval, classification, and string-matching baselines.

Analysis of the 21st Century Cures Act on reducing clinician burden attributable to health information technology and electronic health records 2019 – 2020

Mandatory antibiotic order-associated treatment indications 2017 – 2018

Renal dosing decision support post-electronic medical record implementation 2014 – 2015

Identification of barriers associated with timeliness of antibiotic administration in febrile pediatric patients presenting to the emergency department 2013 – 2014

Evaluation of unfractionated heparin utilization 2013 – 2014

Evaluation of Saccharomyces boulardii utilization 2012 – 2013

Selected Presentations

1. *Basics of Artificial Intelligence*. California Pharmacists Association; Jan 2026. Regional invited presentation.
2. National workshop contributor. CMS Connectathon; 2026.
3. *Path of the Clinician Engineer*. Clinicians.dev Conference; Nov 2024. Invited presentation.
4. *Basics of Artificial Intelligence*. Artificial Intelligence Institute, American Association of Colleges of Pharmacy; Apr 2024. National invited presentation.
5. *The Future of Pharmacy Interoperability*. Office of the National Coordinator for Health IT Annual Meeting; Dec 2023. National panel presentation.
6. *The Evolving Role of Pharmacy in Data and Digital Health*. RPhAlly Soiree; Jul 2023. Invited presentation.
7. *The Informatics Pharmacist*. European Pharmaceutical Students' Association; Jul 2023. International invited presentation.
8. International invited presentation. National Pharmacy Summit Nigeria, Pharma-Tether; Jun 2023.
9. *Building Your Public Health Informatics Toolkit*. American Nursing Informatics Association Midwest; Jun 2022. Regional invited presentation.
10. *Public Health Informatics: Antimicrobial Stewardship and Infection Control*. American Public Health Association Midyear Meeting; Jun 2021. National invited presentation.
11. *Conversion of a Complex Legacy Antimicrobial Stewardship Clinical Decision Support System into Epic*. American Medical Informatics Association Annual Symposium; Nov 2019. National poster presentation.
12. *YouTube, Twitter, and LinkedIn: Integrating Social Media into our Professional Lives*. Minnesota Society of Health-System Pharmacists; Sep 2019. Regional invited presentation.
13. *Boring is the New Goal: Technology Implementation Keys to Success*. ASHP Midyear Clinical Meeting; Dec 2018. National podium presentation.
14. *Enhancing Quality of Care: Pharmacist Clinical Documentation in an Integrated EMR System*. ASHP Midyear Clinical Meeting; Dec 2017. National podium presentation.
15. *Renal Dosing Decision Support Post-Electronic Medical Record Implementation*. Mountain States Conference; Apr 2015. Regional podium presentation.
16. *Optimizing Fate: Pharmacogenomics in Clinical Practice*. University of Utah Health; Mar 2015. Continuing education presentation (ACPE).
17. *Meaningful or Meaningless Use: Are We Doing This Right?* University of Utah Health; Nov 2014. In-service presentation.
18. *Smarter Healthcare: Leveraging Pharmacy Informatics in the Emergency Department*. Sarasota Memorial Hospital; Jun 2014. Lecture.
19. *Identification of Barriers Associated with Timeliness of Antibiotic Administration in Febrile Pediatric Patients Presenting to the Emergency Department*. Florida Residency Conference; May 2014. Regional podium presentation.
20. *Intranasal Mupirocin in the Prevention of Surgical Site Infections in Open-Heart Patients*. Sarasota Memorial Hospital; Jan 2014. In-service presentation.
21. *Nightmare Bacteria: Optimizing Treatment of Multidrug Resistant Gram-Negative Infections*. Sarasota Memorial Hospital; Oct 2013. Continuing education presentation (ACPE).
22. *Good Things Come in Threes: The Role of Apixaban for Atrial Fibrillation Post-PCI*. Sarasota Memorial Hospital; Aug 2013. Continuing education presentation (ANCC).
23. *Neurosyphilis: A Blinding Discovery*. Shands Jacksonville; Feb 2013. Lecture.
24. *The Role of IVIG in Group A Streptococcal Toxic Shock Syndrome*. Sarasota Memorial Hospital; Jun 2012. Lecture.

Teaching

Founder and Instructor 2023 – Present
OpenClinTech
Nine self-paced courses and 137 lessons on AI, APIs, interoperability, and software development for clinicians entering engineering, now extended with live cohort instruction.

Guest Lecturer — ENP 109, Medical Fundamentals 2022 – Present
Tufts University, Engineering Psychology and Human Factors Engineering

Guest Lecturer — ME 250.778, Implementing Fast Healthcare Interoperability Resources Feb 2026, Feb 2025
Johns Hopkins University School of Medicine

Guest Lecturer — ME 250.955, Applied Clinical Informatics Mar 2025
Johns Hopkins University School of Medicine

Preceptor, Clinical Informatics Rotation at Verily 2024 – 2025
PGY-2 Pharmacy Informatics Residents, Johns Hopkins Hospital

Preceptor, Clinical Informatics Capstone at Verily 2024 – 2025
Health Informatics MS Students, University of New England

Guest Lecturer — HLTH_SCI 365-DL, Data Management and Analytics in Healthcare Nov 2024
Northwestern University, Health Sciences

Guest Lecturer — DME100, Essentials of Clinical Data Management and Electronic Data Capture Nov 2024
Seneca Polytechnic

Guest Lecturer — PCS 8040, Pharmacy Management Sep 2022
High Point University, Fred Wilson School of Pharmacy

Guest Lecturer — PHR 8120, Information Systems for Health System Pharmacy Mar 2018
The Ohio State University Wexner Medical Center

Professional Service & Leadership

Pledgee Jan 2026 – Present
CMS Health Tech Ecosystem
Pledged the Kill the Clipboard and Conversational AI use cases, both of which reached general availability, and the Year 2 advanced scheduling and consumer real-time benefits categories.

Panelist, Pharmacy Forecast 2019 – Present
ASHP Foundation

Task Force Member, Policy and Advocacy Workstream 2023 – 2025
American Medical Informatics Association 25x5 Task Force

Committee Member, Value Sets 2022 – 2025
Pharmacy HIT Collaborative

Research Mentor, Computer Science Mentorship Program 2023
Google

Committee Member, Social Media 2018 – 2020
Society of Infectious Diseases Pharmacists

Member, Section Advisory Group for Pharmacy Informatics and Technology on Professional Development 2016 – 2020
American Society of Health-System Pharmacists

Lead Author, FHIR Implementation Guide
HL7 SMART Scheduling Links
Workgroup contributor. Translated multi-organization scheduling workflows into a balloted FHIR implementation guide with examples and developer-facing artifacts.

Honors & Awards

General Availability Recognition, Health Tech Ecosystem Jul 2026
Centers for Medicare & Medicaid Services
Awarded to haau3 for reaching general availability as a patient-facing app in the Conversational AI and Kill the Clipboard categories.

Early Adopter Recognition, Health Tech Ecosystem Jul 2026
Centers for Medicare & Medicaid Services
Awarded for pledging two Year 2 categories, advanced scheduling and consumer real-time benefits.

Verily Values Award, Respect Individuals Dec 2024
Verily

LinkedIn Top Voice Nov 2022
LinkedIn

Top 15 Healthcare Creators Jan 2022
LinkedIn

Creator Accelerator Program Jan 2022
LinkedIn
Inaugural cohort, one of 100 creators selected.

Dean's List 2009 – 2013
University of Florida College of Pharmacy

Phi Kappa Phi Honor Society Dec 2012
Phi Kappa Phi

Rho Chi Honor Society Feb 2011
The Rho Chi Society

Projects & Product Portfolio

Ovant Jul 2026 – Present
Job search product that builds a structured record of a candidate's experience through conversational AI interviews, then writes resumes tailored to a specific role from it, tracks applications, surfaces the keywords a posting rewards, and finds warm referral paths.
Next.js · TypeScript · Postgres · Anthropic API

haau3 platform

May 2026 – Present

B2B platform of microservice APIs that powers haau3 and is available to other organizations. Ships a scheduling slot aggregator conforming to the HL7 SMART Scheduling Links implementation guide, and a FHIR terminology service.

FHIR · SMART Scheduling Links · LOINC · TypeScript

medlistIQ

Apr 2026 – Present

API that embeds informatics pharmacist expertise to turn messy, multi-source FHIR medication data into a clean, inferred medication list.

FHIR · RxNorm · TypeScript

Technology-Enabled Chronic Care

Dec 2025 – Present

Cardio-kidney-metabolic care using FHIR, CDS Hooks, and programmatic terminology services, aligned with CMS ACCESS and FDA TEMPO.

FHIR · CDS Hooks

haau3

Nov 2025 – Present

FHIR-native, AI-powered care management that unifies longitudinal health records so caregivers can focus on the human side of care. Generally available and listed in the CMS Medicare App Library.

Next.js · TypeScript · FHIR · GCP

Cora

Nov 2025 – Present

Component of haau3

Multilingual text and voice conversational AI assistant grounded in patient-authorized records, with source-linked explanations. Deployed independently on GCP and integrated into haau3.

Gemini Live · RAG · TypeScript

Unscripted Rx

Aug 2025 – Present

Pharmacy workforce and recruiting discovery product with 30,000+ connected entities and AI-assisted job-posting ingestion.

Next.js · TypeScript · Vercel

Healthcare Terminology Server

Apr 2025 – Present

Web application for clinicians and informaticists to run terminology server operations in a modern UI, covering RxNorm, SNOMED CT, NPI, and LOINC.

TypeScript · SNOMED CT · LOINC · RxNorm

NPI Directory

Apr 2025 – Present

Lookup for clinicians by National Provider Identifier or name, exposed both as an interactive frontend and as a FHIR terminology service $lookup endpoint.

TypeScript · FHIR

OpenClinTech

2023 – Present

Open and accessible initiative training the next generation of clinician engineers.

Public FHIR Server

Dec 2025

TypeScript and Postgres implementation of core FHIR R4 resource operations, deployed publicly for learning and reference.

TypeScript · Neon Postgres · Vercel

@briankfung/fhir-types

Dec 2025

Hand-rolled FHIR R4 TypeScript types published as a public npm package.

TypeScript

PubMed AI

2024

Retrieval-augmented literature search and summarization over the PubMed corpus, answering a question with cited sources.

Python · RAG

Cardiac 3D Reconstruction

Jun 2023

Python project exploring 3D reconstruction of the heart from imaging modalities, focused on accessible 2D echocardiography images.

Python

Skills

Product	Product strategy, Discovery, Roadmaps, User research, Rapid prototyping, Experimentation, Success metrics, Go-to-market, Product-led growth, Executive communication
AI and machine learning	LLMs, RAG, Conversational and multimodal AI, Model evaluation, Benchmark design, Prompt and model orchestration, Agentic workflows, Synthetic data, Clinical NLP, Fine-tuning
Engineering	Python, TypeScript, SQL, Go, R, Next.js, Tailwind CSS, Prisma, Supabase, FastAPI, PyTorch, REST APIs, GCP, BigQuery, Cloud Healthcare API, Cloud Storage, Cloud Run, Vercel, Docker, OpenTofu, CI/CD, OAuth 2.0
Data and healthcare	Semantic modeling, Knowledge graphs, ETL pipelines, FHIR, SMART on FHIR, SMART Health Links, CDS Hooks, C-CDA, NCPDP SCRIPT, BPM+, CMMN, BPMN, DMN, SNOMED CT, LOINC, RxNorm, UMLS, ICD-10, WHO ATC, NDC
Electronic health records	Epic, Cerner, McKesson, Allscripts